

Pharmacy V13

System Architecture, Management and Decision Intelligence

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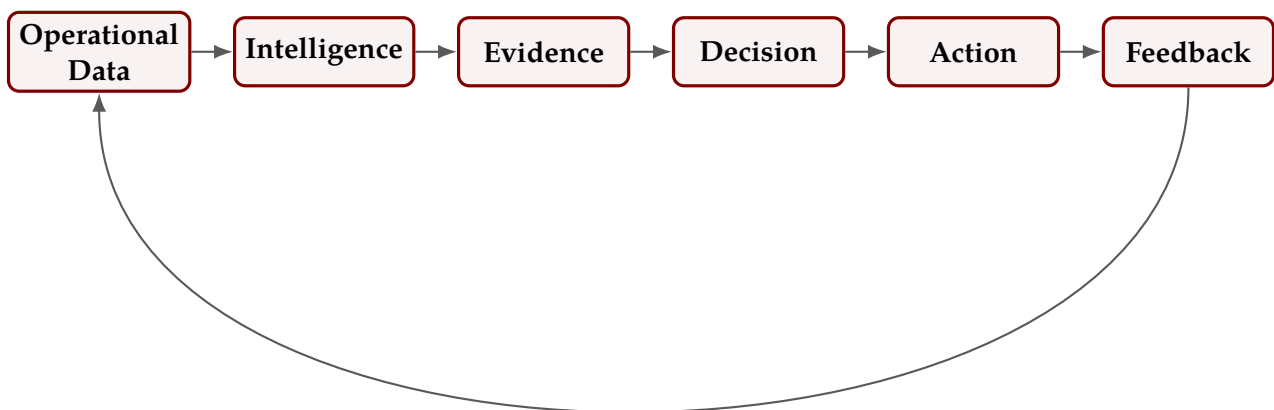
Pharmacy V13 is a management and decision-intelligence system, not merely an inventory application.

The system connects operational pharmacy activity with monitoring, quantitative intelligence, evidence-based decision support, management action, reporting, security, testing, and continuous feedback.

Executive Summary

Pharmacy V13 is an integrated digital pharmacy management platform designed to transform operational activity into structured information, analytical evidence, management decisions, and controlled action.

The central architecture is:



The current production environment is deployed through Vercel with PostgreSQL provided by Neon. Authentication, role-based access, protected APIs, operational transactions, reporting, intelligence services, audit mechanisms, and system-health services form one integrated platform.

The verified live production baseline includes:

- 120 medicines;
- 2 suppliers;
- 7 recorded sales;
- 3 purchase records;
- current Daily Cash revenue of 194;
- authenticated and role-controlled management access;
- protected operational APIs;
- Vercel cloud deployment;
- Neon PostgreSQL connectivity;
- REEM V2 decision-intelligence models;
- reporting and system snapshot functions; and
- **45 PASS / 0 FAIL** in the functional verification suite.

Important data-integrity principle: Live operational data and REEM V2 simulated development data are not interchangeable.

The production database represents actual system transactions. REEM V2 development datasets are used to validate analytical architecture and decision models and must not be presented as live pharmacy observations.

1. System Purpose

The purpose of Pharmacy V13 is to provide a coherent management environment in which pharmacy operations, information, analytics, decisions, and technical infrastructure are connected.

The system addresses six fundamental management questions:

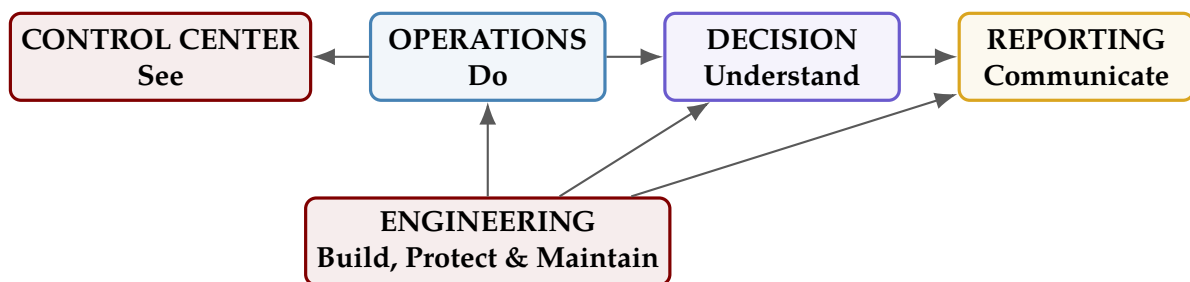
1. What is happening?
2. What has happened?
3. What does the data indicate?
4. What deserves attention?
5. What action should management consider?
6. How can the decision and its outcome be documented and reviewed?

This makes Pharmacy V13 a management information and decision-support platform rather than a collection of isolated screens.

2. Integrated System Architecture

The system is organized into five functional areas:

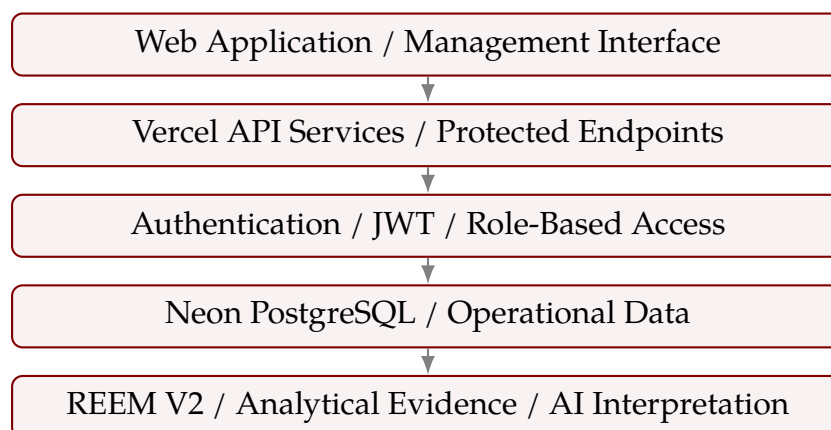
1. **Control Center** — see the current state.
2. **Operations** — perform and record daily activity.
3. **Decision Intelligence** — transform data into analytical evidence.
4. **Reporting** — document and communicate results.
5. **Engineering** — provide security, infrastructure, reliability, testing, and maintenance.



These are different views of one integrated system, not independent applications.

3. Cloud and Technical Architecture

The current production architecture separates the management interface, API services, database, authentication, and analytical intelligence.



Production Components

- **Frontend:** browser-based Pharmacy V13 management interface.
- **Cloud platform:** Vercel production deployment.
- **Database:** Neon PostgreSQL.

- **API layer:** protected Vercel API services.
- **Authentication:** application-managed authentication using secure password hashing and JWT sessions.
- **Authorization:** role-based access control.
- **Analytics:** REEM V2 decision-intelligence layer.
- **Reporting:** controlled system reports, snapshots, and analytical outputs.

4. Authentication and Role-Based Access

Security is part of the system architecture rather than an optional interface feature.

Pharmacy V13 uses application-specific authentication integrated with the production database.

The principal roles are:

Role	Management responsibility
Administrator	Full system administration and privileged operations.
Manager	Operational management and controlled transaction functions.
Analyst	Analysis, intelligence, reporting, and evidence review.
Viewer	Read-only access to authorized information.

Protected APIs require authentication. The health endpoint remains publicly available for service monitoring.

Write operations such as sales and inventory stock transactions are restricted to authorized management roles.

Security principle: Authentication establishes who the user is; authorization determines what that user is permitted to do.

5. Control Center — See

The Control Center provides the management view of the current operational state.

Typical indicators include:

- medicine count;
- sales and revenue;
- inventory quantity and value;
- stock alerts;
- expiry status;

- supplier and procurement status;
- decision-intelligence status;
- daily cash;
- system health; and
- management snapshots.

The Control Center answers:

What is happening in the pharmacy now?

6. Pharmacy Operations — Do

The operational layer records the events that create the pharmacy's digital state.

The main operational entities are:

- medicines;
- suppliers;
- purchases;
- inventory;
- sales;
- stock-in transactions;
- stock-out transactions;
- cash activity; and
- operational alerts.



Every transaction contributes to the evidence available for management and analytical decision-making.

7. Daily Cash

Daily Cash provides a concise financial view of recorded sales.

The current production checkpoint reports:

Indicator	Current value
Recorded sales	7
Revenue	194

The purpose of Daily Cash is not to replace formal accounting. It provides an operational management view derived from system transactions.

8. REEM V2 Decision Intelligence

REEM V2 is the analytical decision-intelligence layer of Pharmacy V13.

Its role is to transform structured operational or development data into analytical evidence that can be interpreted by management and AI.

The principal models are:

1. Demand Forecasting
2. ABC Analysis
3. Expiry Risk
4. Stock Coverage
5. Supplier Analysis
6. Reorder Point
7. Safety Stock
8. Economic Order Quantity

Additional risk analysis includes Monte Carlo simulation where appropriate.

REEM V2 calculates evidence; AI interprets the evidence.

REEM V2 is responsible for quantitative analytical evidence. AI provides a management interpretation of that evidence. The two functions should remain conceptually distinguishable.

9. Forecasting

Demand forecasting estimates future demand from available historical demand information.

The purpose is not simply to produce a number. Forecasting provides a quantitative basis for:

- procurement planning;
- stock planning;
- replenishment decisions;
- capacity planning; and
- identification of changing demand patterns.

Forecasts should always be interpreted together with data quality, observation history, and uncertainty.

10. ABC Analysis

ABC analysis prioritizes medicines according to their relative contribution to the selected value measure.

Its management purpose is to answer:

Which medicines deserve the greatest management attention?

ABC classification can support differentiated control policies, monitoring frequency, procurement attention, and resource allocation.

It is therefore a prioritization tool rather than a simple classification table.

11. Reorder Point and Safety Stock

Reorder Point determines when replenishment should be initiated.

Safety Stock provides protection against uncertainty in demand and replenishment lead time.

Conceptually:

Reorder Point = Expected Lead-Time Demand + Safety Stock

The models therefore connect demand evidence with operational replenishment decisions.

A replenishment signal should be interpreted in relation to current stock, expected demand, lead time, uncertainty, and management policy.

12. Expiry Risk

Expiry analysis identifies medicines and batches that require management attention because of remaining shelf life.

Expiry intelligence can support:

- prioritization of stock;
- FEFO-oriented management;
- reduction of avoidable losses;
- procurement review; and
- early intervention before expiry.

The analytical distinction between batch-level observations and medicine-level summaries must be preserved when interpreting expiry results.

13. Economic Order Quantity

EOQ provides a classical analytical framework for balancing ordering and holding costs under its underlying assumptions.

The conceptual model is:

Optimal Order Quantity balances ordering cost and holding cost.

EOQ is therefore a decision-support model, not an automatic purchasing command.

Management must consider supplier constraints, minimum order quantities, expiry, cash availability, demand uncertainty, and practical procurement conditions.

14. Supplier Intelligence

Supplier intelligence connects procurement records with supplier-level performance information.

Relevant evidence may include:

- purchasing activity;
- supply reliability;
- delivery performance;
- procurement history;
- medicine availability; and

- risk indicators.

The objective is to support supplier evaluation and procurement strategy rather than merely maintain a supplier directory.

15. Monte Carlo Risk Analysis

Monte Carlo analysis provides a mechanism for exploring uncertainty by repeatedly simulating possible outcomes.

In Pharmacy V13, this approach can support questions such as:

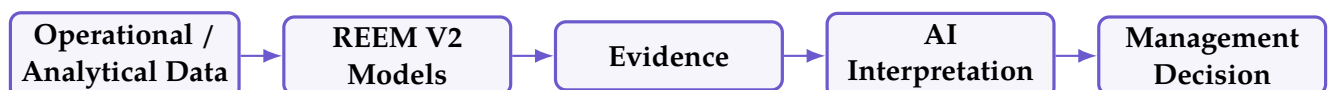
- How sensitive is stock availability to demand variation?
- How much safety stock may be required under uncertainty?
- What is the probability of a stock-out?
- How robust is a replenishment decision?

Simulation outputs are evidence about possible outcomes, not predictions of certainty.

16. AI Management Interpretation

AI is positioned above the analytical evidence rather than replacing the underlying models.

The intended chain is:



This architecture improves transparency because management recommendations can be traced back to analytical evidence.

17. Reporting and Reproducibility

The Report Center provides controlled documentation of the system and its analytical outputs.

Reporting functions include:

- Pharmacy V13 system documentation;
- REEM V2 analytical documentation;
- conceptual architecture documentation;

- management reports;
- SAS-oriented analytical outputs;
- system snapshots;
- system-health information; and
- reproducible source documents.

A report is therefore not simply a printout.

A report is a controlled representation of what the system knows at a particular point in time.

The LaTeX source provides a reproducible documentation layer for the principal system report.

18. Security, Audit and Data Integrity

Pharmacy V13 treats security and data integrity as architectural requirements.

The security architecture includes:

- authenticated access;
- role-based authorization;
- protected operational APIs;
- secure password hashing;
- signed session tokens;
- login-attempt recording;
- application audit records;
- controlled write operations; and
- public health monitoring separated from protected business APIs.

Data integrity also requires a clear distinction between:

1. live operational data;
2. analytical transformations;
3. simulated development datasets;
4. model outputs; and
5. AI interpretations.

No simulated REEM V2 development result should be represented as live pharmacy evidence.

19. Testing and Verification

The Pharmacy V13 functional verification suite currently reports:

45 PASS 0 FAIL

The test suite provides a controlled baseline for subsequent development.

Testing is treated as part of the engineering architecture rather than an activity performed only before release.

Engineering principle: A change is not considered complete merely because the application appears to work. The existing verification baseline must remain intact.

20. Current Production Status

The current Pharmacy V13 production environment is deployed at:

<https://pharmacy-v13.vercel.app/>

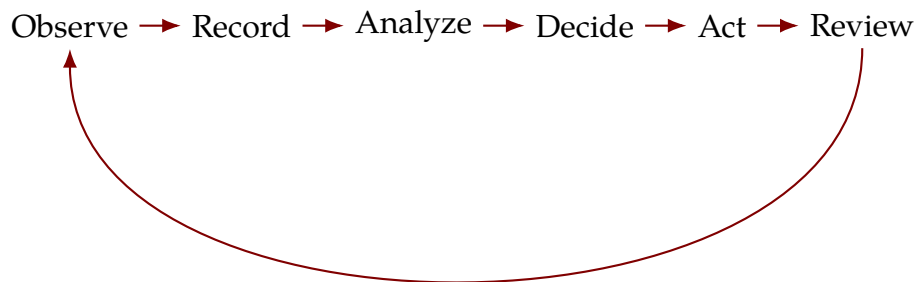
The production health service confirms:

- service status: healthy;
- production API: operational;
- Neon PostgreSQL connection: active;
- database: neondb;
- medicines: 120;
- sales: 7;
- suppliers: 2;
- purchases: 3.

The production application therefore represents a functioning cloud management platform rather than a purely local prototype.

21. Management Decision Cycle

The complete management cycle can be represented as:



This cycle is the conceptual core of Pharmacy V13.

The system observes operational reality, records it, analyzes evidence, supports management decisions, records action, and uses subsequent information for review and improvement.

22. One Integrated Pharmacy Scenario

Consider a medicine approaching its replenishment threshold.

1. A purchase is received from a supplier.
2. The purchase is recorded.
3. Inventory is updated.
4. The Control Center reflects the new operational state.
5. Sales subsequently reduce available stock.
6. Demand evidence is updated.
7. REEM V2 evaluates demand, coverage, reorder point, safety stock, expiry, and related risk.
8. The analytical layer produces evidence.
9. AI interprets the evidence for management.
10. Management decides whether and how to replenish.
11. The resulting action becomes a new operational event.
12. Reporting and snapshots preserve the relevant state for review.

Thus the system closes the loop between operations and decision-making.

23. Live Data versus Development Intelligence

This distinction is essential to the scientific integrity of Pharmacy V13.

Layer	Purpose	Status
Operational PostgreSQL	Real system transactions	LIVE
Authentication and audit	Access and accountability	LIVE
Production APIs	Operational services	LIVE
Control Center	Management interface	LIVE
REEM V2 datasets	Model development and validation	SIMULATED DEVELOPMENT DATA
REEM V2 models	Decision-intelligence architecture	Development / analytical
AI interpretation	Evidence interpretation	Development / analytical

The distinction allows the analytical architecture to evolve without falsely claiming that simulated development evidence represents live pharmacy activity.

24. Engineering Philosophy

Pharmacy V13 follows a deliberately modular architecture.

The system should evolve through controlled increments rather than uncontrolled rewrites.

The principal engineering principles are:

- preserve working functionality;
- make the smallest necessary change;
- protect operational data;
- preserve authentication boundaries;
- maintain test coverage;
- separate operational truth from analytical simulation;
- keep evidence traceable;
- document important changes;
- maintain reproducibility; and
- deploy only verified functionality.

25. Strategic View

Pharmacy V13 can be understood as a small-scale example of a broader digital health and biomedical information architecture.

Its important contribution is not the individual inventory screen or the individual analytical model.

Its value lies in connecting:

Data → Information → Evidence → Decision → Action → Learning

This architecture can support future extensions in supply-chain intelligence, pharmacological information management, biomedical analytics, forecasting, quality assurance, and AI-assisted decision support.

Conclusion

Pharmacy V13 is a unified management, analytical, and engineering platform.

Its architecture connects daily operational transactions with cloud infrastructure, authentication, PostgreSQL data, decision intelligence, AI interpretation, management action, reporting, and feedback.

The fundamental principle is:

Operations create data.
The system organizes the data.
REEM V2 creates analytical evidence.
AI interprets the evidence.
Management makes decisions.
Actions create new operational data.
Reporting preserves and communicates the result.

This closes the management loop and establishes Pharmacy V13 as a practical digital decision-support architecture rather than merely an inventory application.